

Botany short course 2026

Leaves

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What is a leaf?

- Variable in size and shape, form and function

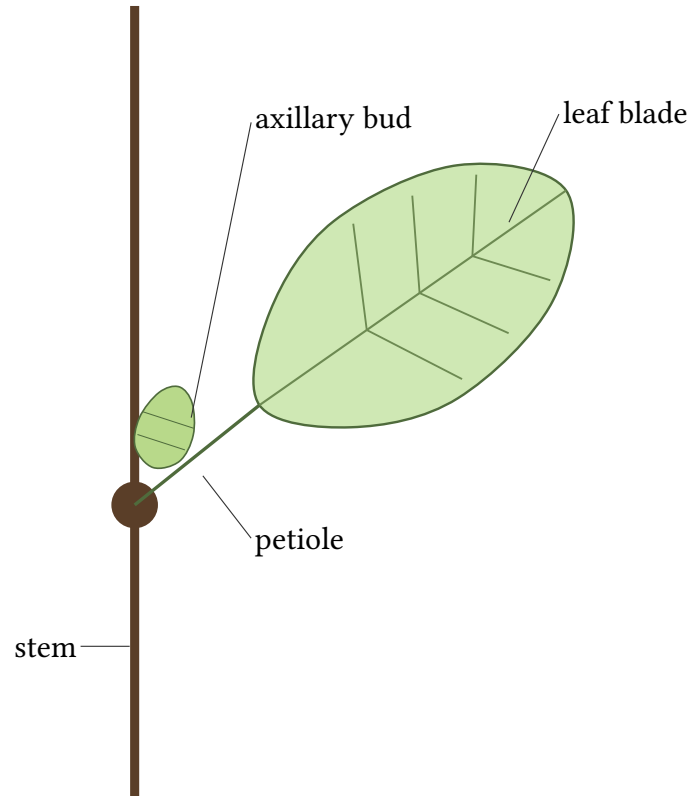
What is a leaf?

- Variable in size and shape, form and function
- Photosynthetic, protective (spines/sheaths), architectural (climbing), carnivorous (pitchers), storage (e.g. garlic)

What is a leaf?

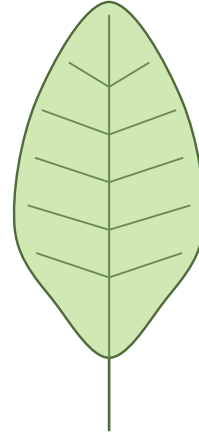
- Variable in size and shape, form and function
- Photosynthetic, protective (spines/sheaths), architectural (climbing), carnivorous (pitchers), storage (e.g. garlic)
- **A leaf always subtends a bud**

Leaves always subtend a bud

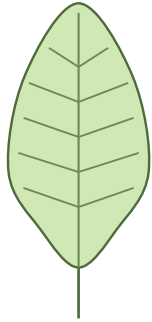


Leaf characters you will classify

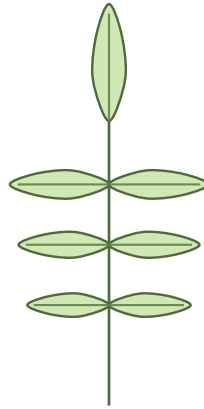
- Simple and compound
- Margins
- Bases
- Overall shape
- Venation
- Any hairs?



Simple vs compound



Simple



Compound: 1-pinnate

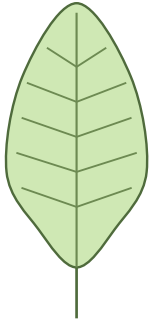


Compound: palmate

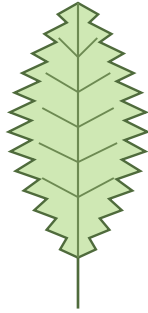
Margins and bases

- Margins look at leaf edges
- Bases are where the petiole joins the leaf

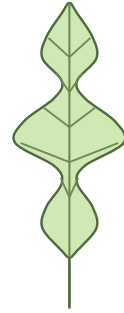
Leaf margins



Entire

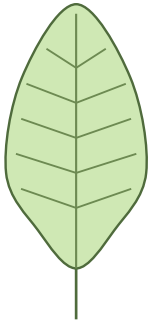


Toothed

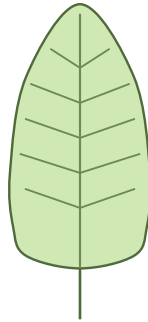


Lobed

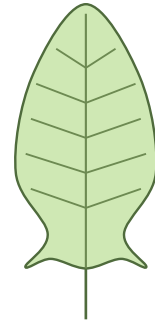
Leaf bases



Cuneate



Rounded



Hastate / sagittate

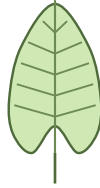
Shape and venation

- Overall leaf shape. There are **many**, we only show a few...
- What do the leaf **veins** look like

Overall leaf shape



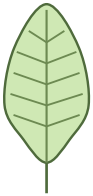
Linear



Cordate



Elliptic



Ovate

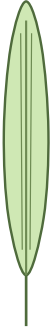


Obovate

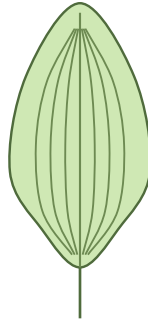


Lanceolate

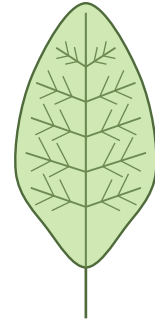
Venation



Parallel (grass-like)



Parallel (*Plantago*)

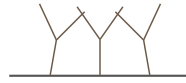


Net-veined

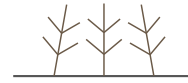
Leaf hairs



Simple hairs (many species)



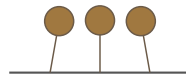
Forked hairs (Brassicaceae)



Dendritic hairs (e.g. *Verbascum*)



Hooked hairs (e.g. *Myosotis*)

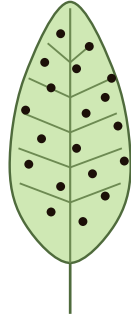


Glandular hairs (many species)

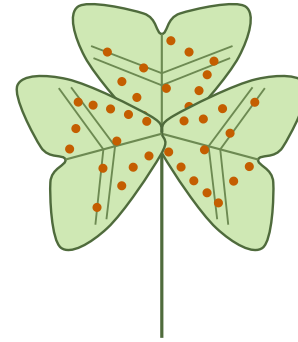


Stellate hairs (e.g. *Hedera*)

Leaf glands



Black scattered glands (e.g. *Hypericum*)



Orange scattered glands (e.g. *Oxalis*)

Activity — Leaf observation

You have a selection of leaves from different species.

1. Pick up one leaf at a time — work through the characters on the next slide
2. Record each on your worksheet
3. Once all leaves described — can you group any together?

Worksheet provided separately

What to record

Leaf characters	
Arrangement	Alternate · opposite · whorled · basal rosette
Simple or compound?	Simple / pinnately compound / palmately compound
Shape	Ovate · lanceolate · linear · oblong · round · lobed
Margin	Entire · toothed (serrate/dentate) · lobed · wavy
Venation	Pinnate · palmate · parallel
Surface	Smooth · hairy · waxy · rough
Notes	Smell? Colour? Petiole? Stipules?